

Curriculum Vitæ

Saurabh Janardan SHIGWAN

1 Personal Details

First Name:	Saurabh
Father Name:	Janardan
Family Name:	SHIGWAN
Date of Birth:	3rd of January, 1989
Place of Birth:	Thane, Maharashtra, India
Citizenship:	Indian
Position:	Assistant Professor
Department:	Department of Computer Science and Engineering (DCSE) Shiv Nadar Institute of Eminence Deemed to be University (SNIOE-DTU), Greater Noida, Uttar Pradesh, India 201314
Email Address:	saurabh.shigwan@snu.edu.in
Homepage:	https://saurabhcsesnu.github.io/my-profile/

2 Academic Career

2.1 Professional Experience

2021–present	Assistant Professor , DCSE, SNIOE-DTU, Greater Noida, Uttar Pradesh, India
2019–2020	Pre-doctorate , Psychiatry Neuroimaging Laboratory (PNL), Harvard Medical School (HMS), Boston, USA
2014–2020	PhD Candidate , DCSE, Indian Institute of Technology Bombay (IIT-Bombay), Mumbai, Maharashtra, India

2.2 Education

2014–2020	PhD , Medical Image Processing, IIT-Bombay, Mumbai, Maharashtra, India Title: “Hierarchical Pointset-Based Statistical Shape Modeling and Applications” Viva: 09/08/2020
-----------	---

	Jury: - Chairman: - Gadre V.M. , Department of Electrical Engineering (DEE), IIT-Bombay, Mumbai, Maharashtra, India - Referees: - Prof. Sylvain Bouix, PNL,HMS, Boston, MA, USA - Prof. Jayathi Sivaswamy, International Institute of Information Technology, Hyderabad (IIIT-Hydrabad), Gachibowli, Hyderabad - Internal Reviewers: - Prof. Ganesh Ramkrishanan, DCSE, IIT-Bombay, Mumbai, India - Advisor: - Prof. Suyash P. Awate, DCSE, IIT-Bombay, Mumbai, India
2012–2014	Master of Technology, Computer Science , Indian Statistical Institute, Kolkata, India
2007–2011	Bachelor of Engineering, Computer Engineering , University of Mumbai, India

3 Research Activities

3.1 PhD Supervision and Co-supervision

2024–present	VSS Tejaswi Abburi —SNIOE-DTU Title: ““Unsupervised Crossing Fiber detection in Multishell Diffusion MRI using Neural Network” Director: Saurabh Shigwan (50%) Co-director: Nitin Kumar (50%) Publications: [CP1] Status: ongoing
2024–present	Vishal —SNIOE-DTU Title: “Applications of Openset Recognition in Medical imaging” Director: Saurabh Shigwan (100%) Status: ongoing
2025–present	Diksha Dhillon —SNIOE-DTU Title: “3D reconstruction from noise prone video sequences” Director: Sumit Shekhar (50%) Co-director: Saurabh Shigwan (50%) Status: ongoing
2021–2024	Abhishek Tiwari —SNIOE-DTU Title: “Deep Learning Based Framework for DTI Parameters Estimation and Analysis for Sparse Diffusion MRI data” Director: Rajeev Kumar (50%) Co-director: Saurabh Shigwan (50%) Publications: [CP8], [J1], [J2], [CP9] Status: completed

3.2 Bachelor’s Students Supervision

2026	Apaar Raina (SNIOE-DTU)
------	-------------------------

2026	“ Sparse view CT Reconstruction” Nithurshen (SNIOE-DTU)
2026	“ Computationally Efficient Segmentation” Arnav Aditya (SNIOE-DTU)
2025	“Openset Recognition in Computer Vision” Mudit Adityaja (SNIOE-DTU)
2024	“Unsupervised Medical Image Segmentation” Bodduluri Saran, Kovvuri Sai Gopal Reddy (SNIOE-DTU)
2023	“Image Segmentation using Graph Neural Networks” Ananya Singhal (SNIOE-DTU)
2023	“DTI parameter estimation using sparse diffusion-MRI measurements” Siddharth Reddy (SNIOE-DTU)
	“Sparse view CT reconstruction”

3.3 Teachings

2026–present	SNIOE-DTU , India Advance Computer vision 22-hour lecture per semester.
2024–2025	SNIOE-DTU , India Introduction to deep learning 22-hour lecture per semester.
2025–present	SNIOE-DTU , India Data Structures and Algorithms 22-hour lectures per semester
2022–2025	SNIOE-DTU , India Digital Image Processing 22-hours lectures per semester
2021–2023	SNIOE-DTU , India Introduction to Probability and Statistics 22-hours lectures per semester

4 Grants and External Funding

4.1 Funding from MoA at Shiv Nadar University (2021-Present)

2026–present	The Habitat Trust (India) Amount awarded: INR 41 Lakh Role: Co-PI [PI: Dr. Sumit Shekhar]
Ecological Conservation	Project Title: “Underwater Video Restoration and 3D Reconstruction for Marine Ecological Conservation”

4.2 Funding during Predoc at HMS (2019–2020)

2016–2018	Brigham and Women’s Hospital (Boston, MA, USA) Amount awarded: \$16,000
-----------	---

Role: Pre-doctoral Fellow

PI: Prof. Yogesh Rathi

Project Title: “Brain Tractography using Diffusion-MRI”

5 Publications

Journal Articles

- [J1] A. Tiwari, S. J. Shigwan, R. K. Singh, et al., “Validation of deep learning techniques for quality augmentation in diffusion mri for clinical studies,” *NeuroImage: Clinical*, 2023, Q1 SCI Journal; Impact Factor = 4.2. DOI: [10.1016/j.nicl.2023.103483](https://doi.org/10.1016/j.nicl.2023.103483).
- [J2] A. Tiwari, R. K. Singh, and S. J. Shigwan, “SwinDTI: Swin transformer-based generalized fast estimation of diffusion tensor parameters from sparse data,” *Neural Computing and Applications*, 2023. DOI: [10.1007/s00521-023-09206-4](https://doi.org/10.1007/s00521-023-09206-4).

Preprints

- [PP1] K. S. G. Reddy, B. Saran, A. M. Adityaja, S. J. Shigwan, and N. Kumar, “Unseggnet: Unsupervised image segmentation using graph neural networks,” *arXiv preprint arXiv:2405.06057*, 2024. [Online]. Available: <https://arxiv.org/pdf/2405.06057>.

Conference Proceedings and Presentations

- [CP1] V. T. Abburi, A. Singhal, S. J. Shigwan, and N. Kumar, “ARMARRecon: An ARMA convolutional filter based graph neural network for neurodegenerative dementias classification,” in *23rd IEEE International Symposium on Biomedical Imaging (ISBI)*, 2026. [Online]. Available: <https://arxiv.org/abs/2601.12067>.
- [CP2] A. Aditya, N. Kumar, and S. J. Shigwan, “UCDSC: Open set uncertainty aware deep simplex classifier for medical image datasets,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026. [Online]. Available: <https://arxiv.org/abs/2511.08196>.
- [CP3] K. Nithurshen and S. J. Shigwan, “Spinecontextresunet: A computationally efficient residual unet for spine ct segmentation,” in *IEEE International Symposium on Computer-Based Medical Systems (CBMS)*, Limassol, Cyprus, 2026. [Online]. Available: <https://arxiv.org/pdf/2605.20760>.
- [CP4] A. M. Adityaja, S. J. Shigwan, and N. Kumar, “UnSegMedGAT: Unsupervised medical image segmentation using graph attention networks clustering,” in *22nd IEEE International Symposium on Biomedical Imaging (ISBI)*, 2025. DOI: [10.1109/ISBI60581.2025.10980693](https://doi.org/10.1109/ISBI60581.2025.10980693).
- [CP5] S. Dubey, S. K. Behera, K. Mittal, M. Ravikiran, N. Kumar, S. J. Shigwan, and R. Saluja, “Multi-feature graph convolution network for hindi OCR verification,” in *1st Workshop on Benchmarks, Harmonization, Annotation, and Standardization for Human-Centric AI in Indian Languages (BHASHA 2025)*, 2025. DOI: [10.18653/v1/2025.bhasha-1.1](https://doi.org/10.18653/v1/2025.bhasha-1.1).
- [CP6] A. Tiwari and S. J. Shigwan, “Tract-specific biomarker discovery for early alzheimer’s disease using sparse diffusion mri and ai framework,” in *International Conference on Pattern Recognition and Machine Intelligence*, Delhi, India, 2025, pp. 369–379. DOI: [10.1007/978-3-032-18480-1_38](https://doi.org/10.1007/978-3-032-18480-1_38).
- [CP7] K. S. G. Reddy, B. Saran, A. M. Adityaja, S. J. Shigwan, N. Kumar, and S. Mukharjee, “UnSeGARmaNet: Unsupervised image segmentation using graph neural networks with convolutional ARMA filters,” in *35th British Machine Vision Conference (BMVC)*, 2024. [Online]. Available: <https://arxiv.org/abs/2410.06114>.

- [CP8] A. Tiwari, A. Singhal, S. J. Shigwan, and R. K. Singh, “Early diagnosis of alzheimer through swin-transformer-based deep learning framework using sparse diffusion measures,” in *The 15th Asian Conference on Machine Learning (ACML 2023)*, 2024. [Online]. Available: <https://proceedings.mlr.press/v222/tiwari24a.html>.
- [CP9] A. Tiwari, A. Singhal, S. J. Shigwan, and R. K. Singh, “Deep learning framework using sparse diffusion mri for diagnosis of frontotemporal dementia,” in *IEEE/CVF International Conference on Computer Vision, BioImage Computing Workshop (ICCV 2023)*, 2023. DOI: [10.1109/ICCVW60793.2023.00413](https://doi.org/10.1109/ICCVW60793.2023.00413).
- [CP10] S. J. Shigwan, A. Gaikwad, and S. P. Awate, “Object segmentation with deep neural nets coupled with a shape prior, when learning from a training set of limited quality and small size,” in *International Symposium on Biomedical Imaging (ISBI-2020)*, Iowa City, USA, 2020. DOI: [10.1109/ISBI45749.2020.9098496](https://doi.org/10.1109/ISBI45749.2020.9098496).
- [CP11] S. J. Shigwan and S. P. Awate, “Hierarchical generative modeling and monte-carlo EM in riemannian shape space for hypothesis testing,” in *Medical Image Computing and Computer Assisted Intervention (MICCAI 2016)*, Athens, Greece, 2016. DOI: [10.1007/978-3-319-46726-9_23](https://doi.org/10.1007/978-3-319-46726-9_23).
- [CP12] A. Gaikwad, S. J. Shigwan, and S. P. Awate, “A statistical model for smooth shapes in kendall shape space,” in *Medical Image Computing and Computer Assisted Intervention (MICCAI 2015)*, Munich, Germany, 2015. DOI: [10.1007/978-3-319-24574-4_75](https://doi.org/10.1007/978-3-319-24574-4_75).